

COURSE PLANNING (RPKPS)

UNIVERSITAS MULTIMEDIA NUSANTARA

VALIDATION PAGE

Course Name : Database Systems
 Course Code : IF351
 Course Coordinator : Vincentius Kurniawan, S. Kom., M.Eng.Sc.
 Team of Lecturers :

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2	YUSTINUS WIDYA WIRATAMA	L00615	
3	NOFRIYADI NURDAM	L00137	
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5	FLORENSIA DAMAYANTI	L01289	

On behalf of the team,
Date: July 2025



Vincentius Kurniawan, S.Kom.,
M.Eng.Sc.
Course Coordinator

Approved by
Date: July 2025

UNIVERSITAS
MULTIMEDIA
NUSANTARA

Arya Wicaksana, S.Kom.,
M.Eng.Sc. (OCA)
Head of Department/Program

Has been checked and considered
comply to UMN standard
Date: July 2025



Assoc. Prof. David Agustriawan, S.Kom.,
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Ex-Officio Informatics Department



COURSE PLANNING (RPKPS) UNIVERSITAS MULTIMEDIA NUSANTARA

COURSE NAME : Database Systems
 CODE / CREDIT : IF351 / 3
 SEMESTER : 3
 PREREQUISITE : IF111
 COURSE STATUS : Mandatory

A. COURSE DESCRIPTION

Students who complete this course will possess a solid understanding of relational models, database design, data normalization, data queries, and database administration using a relational database management system (RDBMS). In addition to the introduction to SQL as Data Definition Language (DDL) and Data Manipulation Language (DML), Transaction Control Language (TCL), managing users and privileges, creating views, stored routines and trigger, database security, Students are able to use advanced SQL function (CTE (common table expression), SQL Window function and analyse time series trend in SQL.

B. LEARNING OUTCOME

B.1. Program Expected Learning Outcomes (ELO) Related to the Course

IQF Level: 6

- ELO 2 Analytical Thinking - Students are able to master basic concepts and theories as well as applied concepts and are able to implement these relationships with an informatics scientific approach and utilize this knowledge to find solutions to problems related to the fields of intelligent systems, information security, game development and software development
- ELO 7 System Administrator - Students are able to manage certain applications or software, including computer systems, database management systems, security systems and intranet systems

B.2. Course Learning Outcomes (CLO)

After passing this course, students will be able to use the skills and knowledge from this course as a beginner level with competences as follow:

- ELO 2 CLO021 Students are able to apply analytical thinking to design and optimize relational and non-relational databases for various applications, including intelligent systems, security, game development, and software development. (C3, C4)
- ELO 2 CLO022 Students are capable to implement and analyse functional dependencies, normalization, and relational database design principles to enhance data integrity and efficiency (C3,C4)

ELO 7	CLO071	Student can manage and secure database systems by implementing SQL commands, transactions, and recovery techniques (C3, C4).
ELO 7	CLO072	Student can implement NoSQL database solutions using MongoDB, focusing on CRUD operations and advanced NoSQL queries (C3, C4).

B.3. Course Sub Learning Outcomes (Sub-CLO)

CLO021	Sub-CLO0211	Students are able to explain the differences between relational and non-relational databases, including analyzing which type of database works best for a given problem (C3, C4).
CLO022	Sub-CLO0221	Students are able to plan database design using Entity-Relationship Diagrams (ERD), apply the normalization process, import datasets, clean large datasets, and organize and transform data by writing SQL queries (C3, C4).
CLO071	Sub-CLO0711	Students are able to perform data selection, manipulation, formatting, aggregation functions, filtering, subqueries, and joins in SQL (C3, C4)
CLO071	Sub-CLO0712	Students are able to apply Transaction Control Language (TCL), manage users and privileges, create views, stored routines, triggers, and implement database security, including advanced SQL functions such as Common Table Expressions (CTE) (C3, C4).
CLO072	Sub-CLO0721	Students are able to perform CRUD operations (Create, Update, Delete) to manage NoSQL databases and use advanced NoSQL queries to optimize data processing (C3, C4).

C. LEARNING ANALYSIS

-Figure is attached-

Several

D. TOPICS

1. Introduction to Database System
2. Functional Dependencies and Normalization for Relational Databases
3. Data Modeling Using the Entity-Relationship Model
4. The Relational Data Model, Relational Database Constraints, and Relational Database Design
5. DDL and DML Statements
6. SELECT Statement 1
7. SELECT Statement 2 and 3
8. Transaction Processing, Concurrency, Recovery Techniques, Database Security
9. Database Security
10. Stored Procedures and Advance SQL Query (CTE)

11. Introduction to NoSQL & MongoDB Fundamentals
12. CRUD Operations & Data Modeling in NoSQL
13. Querying & Indexing for Performance Optimization in MongoDB
14. Group Presentation

E. EVALUATION

1. Attending lectures punctually is mandatory. Students will be considered absent if coming over the specified time.
2. Attending 14 lectures is mandatory. Attending a minimum of 11 from 14 meetings is required to be able to take the final test.
3. Final grade is determined by following components:
 - a. Midterm Test : 30% (20,1% Theory & 9,9% Practicum)
 - b. Final Test : 30% (20,1% Theory & 9,9% Practicum)
 - c. Assignment & Activity : 40% (26,8% Theory & 13,2% Practicum)
 - Active participation : 10%
 - Group Presentation : 20%
 - Quizzes : 10%

Table: Assessment Distribution of Learning Outcomes (Theory)

ELO	CLO	Sub-CLO	Activity (40%)	Midterm (30%)	Final (30%)
ELO 2	CLO021	Sub-CLO0211	5%	5%	-
ELO 2	CLO022	Sub-CLO0221	10%	15%	-
ELO 7	CLO071	Sub-CLO0711	10%	10%	-
ELO 7	CLO071	Sub-CLO0712	10%	-	15%
ELO 7	CLO072	Sub-CLO0721	5%	-	15%

Table: Assessment Distribution of Learning Outcomes (Practicum)

ELO	CLO	Sub-CLO	Activity (40%)	Midterm (30%)	Final (30%)
ELO 2	CLO021	Sub-CLO0211	5%	5%	-
ELO 2	CLO022	Sub-CLO0221	10%	15%	-
ELO 7	CLO071	Sub-CLO0711	10%	10%	-
ELO 7	CLO071	Sub-CLO0712	10%	-	15%
ELO 7	CLO072	Sub-CLO0721	5%	-	15%

FINAL GRADING:

Score	Alphabetical Grade	Numerical Grade	Remarks
85 – 100	A	4	Excellent
80 – 84,99	A-	3,7	Good
75 – 79,99	B+	3,3	
70 – 74,99	B	3,0	
65 – 69,99	B-	2,7	Satisfactory
60 – 64,99	C+	2,3	
55 – 59,99	C	2,0	
45 – 54,99	D	1,0	Poor
0 – 44,99	E	0	Very Poor
	F	0	Academic Violation

F. REFERENCE AND RESOURCES**-Main-**

1. Elmasri, Ramez and Shamkant B. Navathe (2016), *Fundamentals of Database Systems*, 7th edition, Addison Wesley

-Additional/Supporting-

2. Rick F. van der Lans (2007), *SQL for MySQL Developers*, Addison Wesley.
3. Swan, Melanie (2015), *Blockchain*, O'Reilly.
4. <https://learnsql.com/blog/what-is-common-table-expression/> and <https://medium.com/@roshmitadey/common-table-expressions-cte-in-sql-abdf550fc72f>
5. <https://medium.com/@manutej/mastering-sql-window-functions-guide-e6dc17eb1995> and <https://mode.com/sql-tutorial/sql-window-functions>
6. <https://medium.com/@esersaygin/mastering-time-series-analysis-with-sql-d5f8dff8fa7e> and <https://medium.com/analytics-vidhya/advanced-sql-time-series-analysis-fd40eaa08fa3>
7. Bradshaw, Brazil, Chodorow (2019), *MongoDB The Definitive Guide: Powerful and Scalable Data Storage*, O'Reilly
8. Copeland, Rick (2013), *MongoDB Applied Design Patterns*, O'Reilly

G. WEEKLY LESSON PLAN

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	Assessment			Ref.
					Assessment type and Grading System	Indicators	Weight	
1.	Students are able to explain the differences between relational and non-relational databases, including analyzing which type of database works best for a given problem. (Sub-CLO-0211)	<u>Topics:</u> Introduction to Database System <u>Sub-topics:</u> 1. Databases and database users 2. Database system concepts and architectures 3. Distributed database system: <i>Blockchain</i>	<u>Learning Methods:</u> (face to face lecture and practicum) <u>Activities:</u> In class: presentation of teaching materials <u>Task:</u> Individual activities and group activities	2x50' 2x60' 2x60' 170'	<u>Assessment:</u> (Formatif) Active participation Self-paced study and project report and presentation <u>Grading system:</u> Point E	Completeness of self-study and complexity and correctness of the project and active contribution during the class		Book 1: Chapters 1 and 2 Book 3: Chapters 1 and 2
2.	Students are able to plan database design using Entity-Relationship Diagrams (ERD), apply the normalization process, import datasets, clean large datasets, and organize and transform data by writing SQL queries. (Sub-CLO-0221)	<u>Topics:</u> Functional Dependencies and Normalization for Relational Database <u>Sub-topics:</u> Functional dependencies and normal forms based on primary keys	<u>Learning Methods:</u> (face to face lecture and practicum) <u>Activities:</u> In class: presentation of teaching materials <u>Task:</u> Individual activities and group activities	2x50' 2x60' 2x60' 170'	<u>Assessment:</u> Active participation: (Formatif) Self-paced study and project report and presentation <u>Grading system:</u> Point E	Completeness of self-study and complexity and correctness of the project and active contribution during the class		Book 1: Chapter 14
3.	Students are able to plan database design using Entity-Relationship Diagrams (ERD), apply the	<u>Topics:</u> Data Modeling Using the Entity-Relationship Model	<u>Learning Methods:</u> (face to face lecture and practicum) <u>Activities:</u> In class: presentation of teaching materials	2x50'	<u>Assessment:</u> Active participation (Formatif)	Completeness of self-study and complexity and correctness of the project		Book 1: Chapter 3

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	Assessment			Ref.
					Assessment type and Grading System	Indicators	Weight	
	normalization process, import datasets, clean large datasets, and organize and transform data by writing SQL queries. (Sub-CLO-0221)	<u>Sub-topics:</u> Database design, entities, attributes, relationships, ER diagram notations	<u>Task:</u> Individual activities and group activities	2x60' 2x60' 170'	Self-paced study and project report and presentation <u>Grading system:</u> Point E	and active contribution during the class		
4.	Students are able to plan database design using Entity-Relationship Diagrams (ERD), apply the normalization process, import datasets, clean large datasets, and organize and transform data by writing SQL queries. (Sub-CLO-0221)	<u>Topics:</u> Relational Data Model, Relational Database Constraints, and Relational Database Design <u>Sub-topics:</u> Relational data model, relational database constraints, ER-to-relational mapping algorithm	<u>Learning Methods:</u> (face to face lecture and practicum) <u>Activities:</u> In class: presentation of teaching materials <u>Task:</u> Individual activities and group activities	2x50' 2x60' 2x60' 170'	<u>Assessment:</u> (Formatif) Self-paced study and project report and presentation Active participation <u>Grading system:</u> Point E	Completeness of self-study and complexity and correctness of the project and active contribution during the class Note: the grading only for quiz 1 (correctness of the answer)		Book 1: Chapters 5 and 9
5.	Students are able to perform data selection, manipulation, formatting, aggregation functions, filtering, subqueries, and joins in SQL (Sub-CLO-0711)	<u>Topics:</u> DDL & DML Statements <u>Sub-topics:</u> DDL & DML Statements	<u>Learning Methods:</u> (face to face lecture and practicum) <u>Activities:</u> In class: presentation of teaching materials <u>Task:</u> Individual activities and group activities	2x50' 2x60' 2x60' 170'	<u>Assessment:</u> (Formatif) Self-paced study and project report and presentation Active participation <u>Grading system:</u> Point E	Completeness of self-study and complexity and correctness of the project and active contribution during the class		Book 1: Chapters 6 and 7 Book 2: Chapters 4 and 5

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	Assessment			Ref.
					Assessment type and Grading System	Indicators	Weight	
6.	Students are able to perform data selection, manipulation, formatting, aggregation functions, filtering, subqueries, and joins in SQL (Sub-CLO-0711)	<u>Topics:</u> SELECT Statement 1 <u>Sub-topics:</u> SELECT, FROM, WHERE, operators, aggregation functions	<u>Learning Methods:</u> (face to face lecture and practicum) <u>Activities:</u> In class: presentation of teaching materials <u>Task:</u> Individual activities and group activities <u>Lab</u> In Class: practicum	2x50' 2x60' 2x60' 170'	<u>Assessment:</u> (Formatif) Self-paced study and project report and presentation Active participation <u>Grading system:</u> Point E	Completeness of self-study and complexity and correctness of the project and active contribution during the class		Book 1: Chapters 6 and 7 Book 1: Chapters 6-8
7.	Students are able to perform data selection, manipulation, formatting, aggregation functions, filtering, subqueries, and joins in SQL (Sub-CLO-0711)	<u>Topics:</u> SELECT Statement 2 and 3 <u>Sub-topics:</u> GROUP BY, HAVING, ORDER BY, LIMIT, CASE expression, subquery, JOIN and Set operator	<u>Learning Methods:</u> (face to face lecture and practicum) <u>Activities:</u> In class: presentation of teaching materials <u>Task:</u> Individual activities and group activities	2x50' 2x60' 2x60' 170'	<u>Assessment:</u> (Summative) Quiz1 (HCIA-openGauss V1.0 Training) Self-paced study and project report and presentation Active participation <u>Grading system:</u> Point E	Completeness of self-study and complexity and correctness of the project and active contribution during the class Note: the grading for quiz 2 (correctness of the answer) And Self-study Huawei	5% 5%	Book 1: Chapters 6 and 7 Book 1: Chapters 9-12

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	Assessment			Ref.
					Assessment type and Grading System	Indicators	Weight	
Midterm Test Type or Assignment Methods : (Writing test)							30%	W01-W07
8.	Students are able to perform data selection, manipulation, formatting, aggregation functions, filtering, subqueries, and joins in SQL (Sub-CLO-0711)	<u>Topics:</u> Transaction Processing, Concurrency, and Recovery Techniques <u>Sub-topics:</u> Transactions, concurrency, recovery techniques, transaction processing in MySQL	<u>Learning Methods:</u> (face to face lecture and practicum) <u>Activities:</u> In class: presentation of teaching materials <u>Task:</u> Individual activities and group activities	2x50' 2x60' 2x60' 170'	<u>Assessment:</u> (Formatif) Project report and presentation Active participation <u>Grading system:</u> Point E	Complexity and correctness of the project and active contribution during the class		Book 1: Chapters 20-22 Book 2: Chapter 37
9.	Students are able to apply Transaction Control Language (TCL), manage users and privileges, create views, stored routines, triggers, and implement database security, including advanced SQL functions such as Common Table Expressions (CTE). (Sub-CLO-0712)	<u>Topics:</u> Database Security, Managing Users, Privileges, and Views <u>Sub-topics:</u> Introduction to database security issues, concept of a view, discretionary access control, mandatory access control, adding and removing users.	<u>Learning Methods:</u> (face to face lecture and practicum) <u>Activities:</u> In class: presentation of teaching materials <u>Task:</u> Individual activities and group activities	2x50' 2x60' 2x60' 170'	<u>Assessment:</u> (Formatif) Project report and presentation Active participation <u>Grading system:</u> Point E	Complexity and correctness of the project and active contribution during the class		Book 1: Chapters 7 and 30 Book 2: Chapters 26-28

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	Assessment			Ref.
					Assessment type and Grading System	Indicators	Weight	
		privileges, and views.						
10.	Students are able to apply Transaction Control Language (TCL), manage users and privileges, create views, stored routines, triggers, and implement database security, including advanced SQL functions such as Common Table Expressions (CTE). (Sub-CLO-0712)	Topics: Stored Routines, Trigger, and Advance Query (CTE) Sub-topics: Stored procedures, local variables, flow-control statements, querying data with SELECT INTO, MySQL delimiter	Learning Methods: (face to face lecture and practicum) Activities: In class: presentation of teaching materials Task: Individual activities and group activities	2x50' 2x60' 2x60' 170'	Assessment: (Formatif) Project report and presentation Active participation Grading system: Point E	Complexity and correctness of the project and active contribution during the class		Book 1: Chapter 10 Book 2: Chapter 31 https://learnsql.com/blog/what-is-common-table-expression/ https://medium.com/@roshmitadey/common-table-expressions-cte-in-sql-abdf550fc72f
11.	Students are able to perform CRUD operations (Create, Update, Delete) to manage NoSQL databases and use advanced NoSQL queries to optimize data processing. (Sub-CLO-0721)	Topics: Introduction to NoSQL & MongoDB Fundamental Sub-topics:	Learning Methods: (face to face lecture and practicum) Activities: In class: presentation of teaching materials Task: Individual activities and group activities	2x50' 2x60' 2x60' 170'	Assessment: (Summative) Project report and presentation Active participation Grading system: Point E	Complexity and correctness of the project and active contribution during the class		Book 7: Chapter 1 and 2 Note: the grading only for quiz 3 (correctness of the answer)

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	Assessment			Ref.
					Assessment type and Grading System	Indicators	Weight	
12.	Students are able to perform CRUD operations (Create, Update, Delete) to manage NoSQL databases and use advanced NoSQL queries to optimize data processing. (Sub-CLO-0721)	Topics: CRUD Operations Sub-topics: Data Modelling in NoSQL	Learning Methods: (face to face lecture and practicum) Activities: In class: presentation of teaching materials Task: Individual activities and group activities	2x50' 2x60' 2x60' 170'	Assessment: (Formatif) Project report and presentation Active participation Grading system: Point E	Complexity and correctness of the project and active contribution during the class		Book 7: Chapter 3 and 4
13.	Students are able to perform CRUD operations (Create, Update, Delete) to manage NoSQL databases and use advanced NoSQL queries to optimize data processing. (Sub-CLO-0721)	Topics: Querying & Indexing for Performance Optimization in MongoDB	Learning Methods: (face to face lecture and practicum) Activities: In class: presentation of teaching materials Task: Individual activities and group activities	2x50' 2x60' 2x60' 170'	Assessment: (Summative) Quiz 2 Project report and presentation Active participation Grading system: Point E	Complexity and correctness of the project and active contribution during the class Note: the grading only for quiz 4 (correctness of the answer)		Book 7: Chapter 7
14.	Students are able to explain the differences between relational and non-relational databases, including analyzing which type of database works best for a given problem.	Topics: Group report and Presentation Sub-topics: <i>Project report of the project that has been starting from</i>	Learning Methods: Discussion and presentation Activities: Individual activities and group activities	2x50'	Assessment: (Summative) Project report and presentation	Complexity and correctness of the project and active contribution during the class	20%	W01-W13

Week	Course Sub-Learning Outcomes (Sub-CLO)	Topics & Sub-topics	Learning Methods and Activities	Timing	Assessment			Ref.
					Assessment type and Grading System	Indicators	Weight	
	(Sub-CLO-0211) Students are able to plan database design using Entity-Relationship Diagrams (ERD), apply the normalization process, import datasets, clean large datasets, and organize and transform data by writing SQL queries. (Sub-CLO-0221)	<i>the first week of the course (individual) – use your own style in writing</i> <i>Project presentation about the selected topic in building the database, analysis and evaluation (Group)</i>		2x60' 2x60' 170'	Active participation Grading system: Point E	Note: the grading only for the project		
Final Test Type or Assignment Methods: (writing test)							30%	W08-W14



H. Task/Project Details:**1. Task/Project no: 1; Group activity, Week 1-14 (the presentation, complexity and report will be assessed in week 14)**

Course	: Database Systems	Course Code	: IF351		
Task/Project Name	: Group activities	Weight	: 20%		
Related Sub-CLO	: Sub-CLO 0211, 0221, 0711				
A. Individual Activities					
Description	Students learn to collaborate in a team, designing, analysis and evaluating the database system.				
B. Structured Work/Task					
Assessment Type	Project report and presentation				
Description	Individual report (student is prohibited to use the same report with his/her friend, use your own writing, group presentation in week 14)				
Output and format	Output: Project report and project presentation Format: <u>Pdf, PPT and other required files</u>				
Indicators, Criteria, and Weight	a. Complexity of the project (Weight 10%) b. Project report and presentation (Weight 10%)				
Project/Work Timeline	Total duration:	14 weeks			
Others	-				
References	W01 – W14				

2. Task/Project no: 2; active participation, Week 1-14 (the contribution of students in learning process: asking a question, discussion and etc)

Course	: Database Systems	Course Code	: IF351
Task/Project Name	: Active participation	Weight	: 10%
Related Sub-CLO	: Sub-CLO 0211, 0221, 0711, 0712, 0721		
C. Individual Activities			
Description	Students learn to contribute in learning process by asking, discussion and etc.		
D. Structured Work/Task			
Assessment Type	Active participation		
Description	Active participation of each students from week 1-14		
Output and format	Output: A positive discussion to improve the learning project or teaching material Format: <u>Discussion between student and lecturer</u>		
Indicators, Criteria, and Weight	a. Critical thinking and complexity of the question and discussion (Weight 10%)		
Project/Work Timeline	Total duration:	14 weeks	
Others	-		
References	W01 – W14		

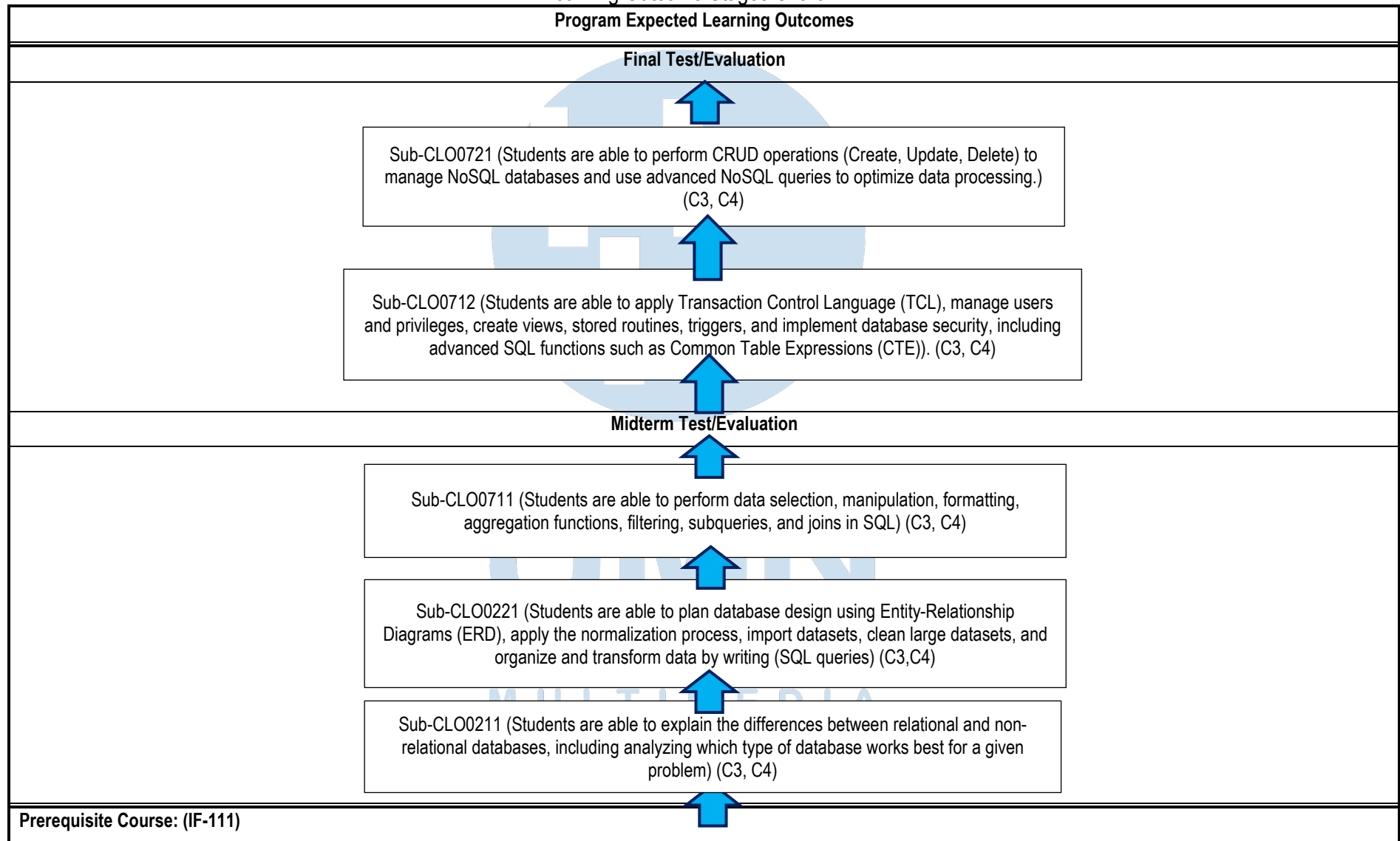


3. Task/Project no:3 Quiz 1-2, Week 7 (HCIA-openGauss v1.0 Training), 14

Course	: Database Systems	Course Code	: IF351
Task/Project Name	: Quiz 1 and 2	Weight	: 5% x 2 (10%)
Related Sub-CLO	: Sub-CLO 0211, 0221, 0711, 0712, 0721		
E. Individual Activities			
Description	Students learn to contribute in learning process by asking, discussion and etc.		
F. Structured Work/Task			
Assessment Type	Question and answer through e-learning		
Description	Each quiz will be tested out of the class schedule for each student to test their individual ability in understanding the delivered materials.		
Output and format	Output: Evaluation on the undemanding of the materials for each student Format: <u>Question and answer in e-learning</u>		
Indicators, Criteria, and Weight	a. Critical thinking and complexity of the question and discussion (Weight 10%) b. Completing the course material with a minimum 10 hours recorded in each student account (Weight 5%) c. Pass the mock up exam (Weight 5%)		
Project/Work Timeline	Total duration:	2 times in 14 weeks	
Others	-		
References	W01 – W014		

Attachment: Learning Outcome Analysis

Learning Outcome Stages Chart



I. Revision History

Course Code	Revision No	Date in Effect	Changes
IF351	5	23 August 2024	Additional materials for advance SQL topic on session 13 and 14 both theory and lab, additional Self-paced study (HCIA-openGauss V1.0 Training), Group project to solve real life problems on session 14.
IF351	6	26 August 2025	Update ELO with the latest curriculum. Adjust additional Self-paced study (HCIA-openGauss V1.0 Training) into Quiz, Additional material for NoSQL and MongoDB on session 11, 12, 13 both theory and lab.

